Selected Questions & Worked Solutions

Question 1

A factory has three boxes containing colored balls. Box A contains 6 red, 4 blue, and 2 green balls; Box B contains 5 red, 7 blue, and 3 green balls; and Box C contains 8 red, 2 blue, and 5 green balls. One box is selected at random, but Box B is twice as likely to be chosen as each of the other boxes. A ball drawn from the chosen box is found to be blue. What is the probability that the selected box was Box C? (5 Marks)

Solution

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Given: P(A) = 1/4, P(B) = 1/2, P(C) = 1/4
P(Blue|A) = 4/12 = 1/3
P(Blue|B) = 7/15
P(Blue|C) = 2/15
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Bayes' Theorem:

$$P(C|Blue) = [P(Blue|C) \times P(C)] / \Sigma[P(Blue|B_i) \times P(B_i)]$$
$$= (2/15 \times 1/4) / [(1/3 \times 1/4) + (7/15 \times 1/2) + (2/15 \times 1/4)]$$
$$= (1/30) / (1/12 + 7/30 + 1/30)$$
$$= (1/30) / (7/20)$$
$$= 2/21 \approx 0.0952$$

Answer = 9.52% (≈ 10%)

Question 4

Examine the performance of a machine-learning model using its training and validation loss values over multiple epochs: (5 Marks)
(a) Epoch 1: Training Loss = 2.8, Validation Loss = 2.4
(b) Epoch 2: Training Loss = 2.5, Validation Loss = 2.2
(c) Epoch 3: Training Loss = 2.2, Validation Loss = 2.0
(d) Epoch 4: Training Loss = 2.0, Validation Loss = 2.1
(e) Epoch 5: Training Loss = 1.8, Validation Loss = 2.3
(f) Epoch 6: Training Loss = 1.6, Validation Loss = 2.6
(g) Epoch 7: Training Loss = 1.5, Validation Loss = 2.9
(h) Epoch 8: Training Loss = 1.3, Validation Loss = 3.2
i) Identify the epoch at which the model begins to overfit.
ii) Suggest two effective techniques to reduce overfitting in this model.
iii) Discuss how overfitting affects the model's generalization ability on unseen data.

Solution

Validation loss decreases until Epoch 3, then increases from Epoch 4 onward while training loss keeps decreasing throughout.

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i) Overfitting starts at Epoch 4.
ii) Techniques: Early stopping, Dropout / L2 regularization.
iii) Overfitting causes the model to memorize training data patterns and
noise rather than learning generalizable features, so it performs well on
training data but poorly on unseen (validation/test) data — reducing
real-world reliability.
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Question 8

In a medical diagnosis system, an AI model detects whether a patient has a disease or not. Out of 300 cases (150 diseased and 150 healthy): (5 Marks)

a) Diseased identified as Diseased: 138

b) Diseased identified as Healthy: 12

c) Healthy identified as Healthy: 135

d) Healthy identified as Diseased: 15

Compute: Accuracy, Precision, Sensitivity (Recall), Specificity, and F1-Score.

Solution

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TP = 138, TN = 135, FP = 15, FN = 12
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Accuracy = (TP+TN)/Total = (138+135)/300 = 91.00%
Precision = TP/(TP+FP) = 138/153 = 90.20%
Recall = TP/(TP+FN) = 138/150 = 92.00%
Specificity = TN/(TN+FP) = 135/150 = 90.00%
F1 = 2 × Precision × Recall / (Precision+Recall) = 91.09%
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The model performs strongly and consistently across all metrics, indicating balanced and reliable diagnostic performance.

Question 11

In Shop X, there are 20 bottles of pure oil and 30 bottles of adulterated oil, while in Shop Y, there are 40 bottles of pure oil and 20 bottles of adulterated oil. A bottle is selected at random from one of the shops, with Shop Y being twice as likely to be chosen as Shop X. The selected bottle is found to be adulterated. (5 Marks)

Solution

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Given: P(X) = 1/3, P(Y) = 2/3 (Shop Y twice as likely as Shop X)
Shop X total = 50, Shop Y total = 60
P(Adulterated|X) = 30/50 = 0.6
P(Adulterated|Y) = 20/60 = 1/3
```

Bayes' Theorem (probability the bottle came from Shop X, given it is adulterated):

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P(X|Adulterated) = [P(Adulterated|X) × P(X)] / [P(Adulterated|X)×P(X) +
P(Adulterated|Y)×P(Y)]
= (0.6 × 1/3) / [(0.6)(1/3) + (1/3)(2/3)]
= 0.2 / (0.2 + 0.2222)
= 0.2 / 0.4222
= 9/19 ≈ 0.4737
```

Answer: P(Shop X | Adulterated) = 9/19 ≈ 47.37%

(Equivalently, P(Shop Y | Adulterated) = 10/19 ≈ 52.63%, since the two probabilities must sum to 1.)

Question 13

In a binary classification problem using a deep learning model with a K-Nearest Neighbor classifier, the model produces True Positive (TP) = 120 and True Negative (TN) = 150. The dataset contains 1000 samples per class, and 20% of the data is used for testing. (5 Marks) Find the following: Accuracy, Precision, Sensitivity (Recall), Specificity.

Solution

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Test samples per class = 1000 × 20% = 200
FN = 200 − TP = 200 − 120 = 80
FP = 200 − TN = 200 − 150 = 50
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Accuracy = (TP+TN)/Total = (120+150)/400 = 67.5%
Precision = TP/(TP+FP) = 120/170 = 70.59%
Recall (Sensitivity) = TP/(TP+FN) = 120/200 = 60%
Specificity = TN/(TN+FP) = 150/200 = 75%
```

Inference: Moderate overall accuracy with noticeably lower recall than specificity, meaning the model misses a fair number of true positive cases.